

Deep Reference Appendix

Complete Clinical Records Reference — 2024–2026

Sarah K. Sahl • DOB: May 18, 1984 • Prepared June 2026

About This Document

This appendix contains the complete raw reference data underlying the Primary Clinical Summary. It is organized for provider lookup rather than narrative reading. All values are sourced directly from Mayo Clinic records (April–May 2026) except where noted. For context and interpretation, refer to the Primary Clinical Summary.

1. Patient Identification & Provider Directory

1.1 Clinical Demographics

Patient Name	Sarah K. Sahl
Date of Birth	May 18, 1984 (Age 41)
Gender	Female
Address	2623 NW Penny Ln, Ankeny, IA 50023-8355
Primary Contact	515-231-3688 (Mobile/Home) misssahl@outlook.com
Language	English (Spoken and Written)
Marital Status	Single
Race/Ethnicity	White / Not Hispanic or Latino
Known Allergies	NKAA — No Known Active Allergies (as of 05/29/2026)
Housing Status	Has current housing; expressed concern about future stability (03/30/2026 social determinants screen)
Social Determinants (03/30/2026)	Hunger, Transportation, and Utilities: “Never true” or “No” risk

1.2 Mayo Clinic Provider Directory (April–May 2026)

Provider	Specialty / Role	Clinical Responsibility
Dr. Maria Poirier	Internal Medicine / Pulmonology	Authorizing Provider: PFTs, CT Chest, Genetic consultation order
Dr. Arushi Khurana	Hematology / Oncology	Authorizing Provider: Hematology, FISH, Immunology
Dr. Naoki Takahashi	Urology / Radiology	Interpretive Radiologist: CT Urogram
Dr. Allison Ferreira	Radiology	Interpretive Radiologist: CT Chest, DX Chest

Dr. Adam Johnson	Musculoskeletal Radiology	Interpretive Radiologist: Outside MRI (10/20/2023)
Dr. Horatiu Olteanu	Hematopathology	Reviewer: Leukemia/Lymphoma Flow Cytometry
Dr. Aruna Rangan	Hematopathology	Signing Pathologist: T-Cell TCR Gene Rearrangement
Dr. Jess F. Peterson	Cytogenetics	Reviewing Pathologist: Chronic Eosinophilia FISH

1.3 Chronological Encounter Log — April–May 2026

Date	Category	Activity / Result
04/01/2026	Imaging & ECG	DX Chest (2-view); CT Urogram (with contrast); ECG
04/01/2026	Laboratory	Urinalysis and Bacterial Culture Finalized
04/03/2026	Intervention	Urology Cystoscopy finalized
04/08/2026	Vitals	BP 109/75, Weight 67.4 kg
04/09/2026	Laboratory	CBC, Morphology, Autoimmune Serology (ANA, ANCA, Tryptase)
04/09/2026	Imaging Review	Interpretation of 10/20/2023 outside pelvic MRI (Dr. Adam Johnson)
04/10/2026	Laboratory	Leukemia/Lymphoma Phenotype (Flow Cytometry) Finalized
04/13/2026	Laboratory	T-Cell Receptor Gene Rearrangement (PCR) Finalized — Negative
04/14/2026	Laboratory	Chronic Eosinophilia Diagnostic FISH Finalized — Normal all loci
04/15/2026	Diagnostics	PFT; FeNO; CT Chest (without contrast); GI Pathogen Panel; O&P Microscopy
04/16/2026	Intervention	Mepolizumab (Nucala) 100mg SQ — initiated
04/16/2026	Vitals	Pulse 74 bpm, SpO2 98%
04/17/2026	Laboratory	O&P Microscopy finalized — negative
05/01/2026	Administrative	Active Problem List updated: Trigonitis Bladder added
05/29/2026	Record Review	Comprehensive Patient Health Summary generated

2. Pulmonary & Respiratory Diagnostics

2.1 Complete PFT Data — 04/15/2026 (1:54 PM CDT)

Component	Pre-Bronchodilator	Post-Bronchodilator (Albuterol)	Clinical Note
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FVC (L)	3.65	4.06	+11% improvement
FEV1 (L)	2.10	2.99	+42% improvement — significant
FEV1/FVC (%)	57.39%	73.70%	Obstructive pre; +16 pts post
FEF 25-75% (L/s)	1.00	2.21	+121% — major small airways response
FEF 50/FIF 50 (%)	31.31%	62.02%	
PEF (L/s)	5.17	7.12	
PIF (L/s)	5.50	5.32	
TLC (L)	5.83	—	Air trapping context
RV (L)	2.13	—	Residual volume
DLCO (ml/min/mmHg)	17.68	—	
DLCOc (ml/min/mmHg)	18.33	—	Corrected DLCO

2.2 Exhaled Nitric Oxide (FeNO)

Test	Value	Reference / Interpretation
FeNO (Exhaled NO Oral)	82 ppb — FLAGGED HIGH	ULN: 39 ppb. More than 2× upper limit. Strongly correlates with eosinophilic airway inflammation (Th2-high phenotype). Clinical rationale for mepolizumab initiation.
Concurrent medications at test time	Albuterol, Advair (fluticasone-salmeterol), Singulair (montelukast)	All listed were active at time of FeNO measurement

2.3 CT Chest Without Contrast — 04/15/2026 (Dr. Allison Ferreira, Radiology)

- **Findings:** Prominence of right paratracheal stripe (likely summation artifact). Mild diffuse bronchial wall thickening. Mild patchy air trapping bilaterally. Few right apical blebs. Accessory fissure in right lower lobe. 9 mm sclerotic focus in T3 vertebral body.
- **Impression:** Small marginal airways disease with patchy air trapping and small airways obstruction (nonspecific).

2.4 DX Chest 2-View — 04/01/2026

- Findings: Mild prominence of the right paratracheal stripe
- Impression: CT chest recommended to evaluate paratracheal stripe prominence — CT performed 04/15/2026

3. Hematology, Immunology & Genetic Testing

3.1 Complete Blood Count — 04/09/2026

Component	Value	Reference Range	Status
Hemoglobin	12.3 g/dL	11.6–15.0	Normal
Leukocytes (WBC)	4.4 x10(9)/L	3.4–9.6	Normal
Neutrophils (automated)	1.51 x10(9)/L	1.56–6.45	Low (L)
Neutrophils (manual ANC)	1.85 x10(9)/L	1.56–6.45	Normal
Lymphocytes	1.90 x10(9)/L	0.95–3.07	Normal
Monocytes	0.32 x10(9)/L	0.26–0.81	Normal
Eosinophils (automated)	0.58 x10(9)/L	0.03–0.48	HIGH (H)
Basophils (automated)	0.06 x10(9)/L	0.01–0.08	Normal

3.2 Manual Morphology / Special Smear — 04/09/2026

Component	Manual %	Reference Range	Status
Neutrophilic Segs and Bands	42%	50–75%	Low
Eosinophils	12%	1–3%	HIGH (H)
Basophils	3%	0–2%	HIGH (H) on %
Manual Absolute Neutrophil Count	1.85 x10(9)/L	1.56–6.45	Normal

Neutrophil discrepancy: Automated count 1.51 Low vs manual ANC 1.85 Normal. In the setting of active eosinophilia, manual differential is more reliable. The discrepancy does not represent a clinical neutropenia.

3.3 Autoimmune & Inflammatory Serology — April 2026

Test	Value	Interpretation
Antinuclear AB (ANA)	<1:80	Negative — autoimmune CTD excluded
ANCA (Myeloperoxidase Ab)	<0.2	Negative — ANCA-positive vasculitis excluded
ANCA (Proteinase 3 Ab)	<0.2	Negative
Tryptase, Serum	2.3 ng/mL	Normal (ref <11.5) — systemic mastocytosis excluded
Rheumatoid Factor (RF)	17.3 U/mL	Elevated — anti-CCP testing outstanding to clarify significance

IgE	42.4 IU/mL	Normal (ref ≤214)
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3.4 Chronic Eosinophilia Diagnostic FISH — 04/14/2026

- **Result: Normal — No evidence of common eosinophilia-associated rearrangements**
- Reviewed by: Dr. Jess F. Peterson, Cytogenetics

Loci studied (all normal):

Locus	Gene / Marker	Clinical Significance
4q12	CHIC2 deletion	FIP1L1-PDGFR fusion — chronic eosinophilic leukemia
5q32	PDGFRB sep	PDGFRB rearrangement — myeloid neoplasm
9p24.1	JAK2 sep	JAK2 rearrangement
9q34	ABL1 sep	ABL1 rearrangement
13q12.2	FLT3 sep	FLT3 rearrangement
8p11.2	FGFR1 sep	FGFR1 rearrangement — stem cell myeloproliferative disorder

3.5 Flow Cytometry & TCR Gene Rearrangement

Test	Date	Result	Reviewer
Leukemia/Lymphoma Phenotype (Flow Cytometry)	04/10/2026	Normal — no monocytic B-cells, no aberrant T-cell populations. TRBC1 and TCR-gamma/delta normal phenotype.	Dr. Horatiu Olteanu, Hematopathology
T-Cell Receptor Gene Rearrangement (PCR)	04/13/2026	Negative — no clonal T-cell receptor gene rearrangement detected	Dr. Aruna Rangan, Hematopathology

3.6 GI & Infectious Pathogen Screening — 04/15–17/2026

- GI Pathogen Panel (PCR, 04/15/2026): Negative for all bacterial, viral, and parasitic targets
- Ova and Parasite Microscopy (finalized 04/17/2026): No parasites seen. Leukocytes present.

Note: Single negative O&P does not rule out infection per clinical note. No clinical concern raised at current evaluation stage.

4. Cardiology & Cardiovascular Diagnostics

4.1 ECG — 04/01/2026

Parameter	Value	Reference / Note
Ventricular Rate	59 BPM	Normal
PR Interval	148 ms	Normal
QRSD Interval	78 ms	Normal
QT Interval	416 ms	
QTc Interval	411 ms	Normal

4.2 NT-proBNP Longitudinal Reference

Date	Value	Reference	Context
04/01/2026	263 pg/mL (H)	≤162 pg/mL	Mayo Clinic. Flagged abnormal. Below acute HF triage threshold (300) but above chronic cardiomyopathy monitoring ref.
02/08/2026	4 pg/mL (Normal)	≤162 pg/mL	Broadlawns ED. No prior baseline in record for trending.

4.3 Echocardiogram Reference (07/2024)

- Ejection Fraction trend: 56% → ~50% (declining)
- Redundant interatrial septum — tissue laxity marker, associated with hypermobility disorders
- Mild mitral valve regurgitation
- Mild tricuspid valve regurgitation
- Mild aortic valve regurgitation
- Cardiomyopathy type: Nonischemic, dilated pattern (TTN pathogenic variant)

Source file: 20240701_echocardiogram.pdf (maintained in longitudinal record)

5. Urology & Internal Medicine

5.1 CT Urogram — 04/01/2026 (Dr. Naoki Takahashi, Radiology)

- **Finding:** 3 mm filling defect with probable enhancement in posterior base of bladder
- **Longitudinal comparison:** Stable and unchanged vs 10/13/2016 and 12/23/2025
- **Impression:** Requires correlation with cystoscopy (performed 04/03/2026). No urachal remnant or cyst.
- **Ancillary:** Tiny hepatic cysts/hemangiomas; splenule identified

5.2 Urinalysis — 04/01/2026

Component	Value	Interpretation
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Glucose	Negative	Normal
WBC (microscopy)	1-3 /hpf	Automated flag “Abnormal” — within female-specific reference range (<11 /hpf). Clinically normal.
Bacterial Culture	Urogenital microbiota	Susceptibilities not performed. No pathogen isolated.
Predicted 24hr Protein	224 mg/day	Borderline (ref <229). Monitoring warranted in eosinophilic disease context.

5.3 Vital Signs Reference

Date	Vital	Value
04/08/2026	Blood Pressure	109/75 mmHg Weight: 67.4 kg
04/16/2026	Pulse / SpO2	74 bpm 98%
Separate encounter	Diastolic HTN documented	98 mmHg diastolic — persisting on metoprolol 100mg BID (maximum beta blockade)

6. Musculoskeletal & Pelvic Imaging

6.1 Outside Pelvic MRI Interpretation — 04/09/2026 (Dr. Adam Johnson, MSK Radiology)

Original study date: 10/20/2023. Reviewed and interpreted by Mayo Clinic 04/09/2026.

Structure	Finding
Sacral / pelvic	No MRI findings to suggest acute or chronic sacral otitis
Anterior superior labrum	Degenerative blunting/irregularity
Gluteal tendons	Intact without significant tendinosis (Mayo re-read). Note: prior outside reads documented partial gluteal tendon tearing — discrepancy between reads of same imaging.
Greater trochanteric bursae	Trace edema/inflammation present (confirmed both reads)

MRI read discrepancy: Prior outside radiology documented partial gluteal tendon tearing. Mayo MSK radiology (04/09/2026) re-read of the same 10/20/2023 study describes tendons as intact, with trace bursal edema. Greater trochanteric bursal inflammation was confirmed by both reads. Discrepancy is noted and documented.

7. Complete Medication Reference

Meloxicam: DISCONTINUED as of June 2026. Listed as active in 05/29/2026 health summary — this is a records lag. NSAIDs strictly contraindicated (cardiac + renal). Records conflict acknowledged.

Medication	Dose/Route	Frequency	Started	Status / Notes
Mepolizumab (Nucala)	100mg/mL SQ	Monthly	04/16/2026	ACTIVE — NEW. Anti-IL-5 biologic.
Metoprolol succinate	100mg Oral	BID	01/03/2026	ACTIVE. Beta-1 blocker. ADRB1 PGx variant.
Fluticasone-salmeterol (Advair)	250-50mcg Inhaled	1 puff BID	02/08/2026	ACTIVE. ADRB2/GLCCI1 PGx variants.
Gabapentin (Neurontin)	400mg Oral	4x daily	01/07/2020	ACTIVE — UNDER REVIEW 06/18/2026. Pregabalin being considered.
Baclofen (Lioresal)	10mg Oral	Daily PRN	03/09/2020	ACTIVE — UNDER REVIEW 06/18/2026. For spasms.
Furosemide	20mg Oral	PRN	07/08/2024	ACTIVE. Refill pending June 2026.
Potassium Chloride	10 mEq Oral	With furosemide	07/08/2024	ACTIVE.
Tramadol (Ultram)	50mg Oral	q6h PRN	02/19/2018	ACTIVE PRN. OPRM1/CYP2D6 PGx variants.
Rosuvastatin (Crestor)	5mg Oral	Daily	12/05/2025	ACTIVE. SLCO1B1 myopathy risk PGx.
Fluoxetine (Prozac)	10mg Oral	Daily	02/26/2026	ACTIVE. CYP2C19 poor metabolizer.
Buspirone (BuSpar)	5mg Oral	BID	03/14/2026	ACTIVE.
Alprazolam XR (Xanax XR)	0.5mg Oral	Daily	—	ACTIVE.
Montelukast (Singulair)	10mg Oral	Daily (bedtime)	09/10/2024	ACTIVE.
Albuterol	90mcg/act 2 puffs	PRN	—	ACTIVE. Rescue bronchodilator.
Estradiol (Climara)	0.1mg/24hr patch	2x weekly	—	ACTIVE. Must Rx as 1-month supply (Iowa Total Care). Amazon Pharmacy.
Cetirizine (Zyrtec)	10mg Oral	Daily	—	ACTIVE. Antihistamine.
Atomoxetine (Strattera)	Active	—	—	ACTIVE. ADHD. CYP2D6 PGx.

Meloxicam (Mobic)	7.5mg Oral	Daily	11/07/2022	DISCONTINUED June 2026. NSAID — strictly contraindicated.
Acetaminophen	1,000mg Oral	PRN	—	ACTIVE — current primary analgesic. Inadequate pain control.
Prednisone	20mg Oral	Daily PRN	09/30/2022	PRN only. Steroid resistance documented.

8. Longitudinal Problem List

Diagnosis	Noted	System
Cardiomyopathy in Diseases Classified Elsewhere (TTN pathogenic variant)	04/01/2026	Cardiac
Cardiomyopathy Family History (father deceased, brother progressing)	04/01/2026	Cardiac / Genetic
Eosinophilia Unspecified	04/01/2026	Hematologic
Asthma Severe Persistent (eosinophilic phenotype)	04/01/2026	Pulmonary
Hypermobility Joint (Beighton markers confirmed)	04/01/2026	Musculoskeletal
Chronic Pain Syndrome	04/01/2026	Pain
Posttraumatic Stress Disorder, Prolonged	04/01/2026	Psychiatric
Allergy Pollen	04/01/2026	Allergic
Tachycardia / Inappropriate Sinus Tachycardia	04/01/2026	Cardiac
Genetic Carrier of Other Disease	04/01/2026	Genetic
Mass Bladder (3mm, stable since 2016)	04/01/2026	Urologic
Trigonitis Bladder	05/01/2026	Urologic
OCD	Active	Psychiatric
ADHD (on Atomoxetine)	Active	Psychiatric / Neurodevelopmental
Idiopathic Urticaria	Active (since ≥2011)	Dermatologic / Allergic